

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Notary Document Automation and Office Management System (NDAOMS)

Version:			1.0
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Section 1 – Introduction

1.1 Purpose

The application is designed to digitize and automate the internal document preparation process of a notary advocate office. It replaces manual record-keeping, repetitive document drafting, and paper-based client management with a centralized web-based solution. The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements for the Notary Document Automation & Office Management System.

The software stores client information once and reuses it while generating affidavits and other legal documents from predefined templates. This significantly reduces repetitive typing, minimizes human error, improves office productivity, and maintains permanent client records for future reference.

This SRS serves as the primary reference document for stakeholders, developers, testers, and future maintenance teams.

1.2 Document Purpose

The purpose of this document is to provide a comprehensive specification of the proposed software system before development begins.

Specifically, this document aims to:

- Define the business objectives of the project.
- Describe the overall functionality of the application.
- Specify all functional requirements.
- Define non-functional requirements.
- Describe user roles and responsibilities.
- Document business rules.
- Define database requirements.
- Describe system architecture.
- Provide a reference for software development.
- Provide a reference for software testing.
- Support future maintenance and enhancements.

This document will act as a contractual reference between the client and the development team to ensure that the delivered software satisfies the agreed requirements.

1.3 Project Overview

The **Notary Document Automation and Office Management System** is a secure web-based application developed specifically for notary advocate offices.

The software is intended exclusively for internal office operations and is not designed for direct access by clients or external users.

The application centralizes all major office activities related to legal document preparation. It maintains client records, manages reusable document templates, generates legal documents automatically, stores generated documents for future reference, assigns document preparation tasks to Junior Advocates, and provides administrative reporting.

One of the key objectives of the system is to eliminate repeated data entry. Once a client's information has been entered into the system, the same information can be reused for generating multiple affidavits and legal documents without requiring the user to type the details again.

Every generated document is automatically archived and linked to the corresponding client, enabling advocates to retrieve previously generated documents whenever required.

The application also improves collaboration within the office by allowing the Senior Advocate to distribute document preparation work among Junior Advocates through an integrated task assignment module.

1.4 Scope of the System

The proposed system focuses exclusively on automating the internal operations of a notary advocate office.

The application includes the following functional areas:

Client Management

The system shall maintain a permanent digital repository of all clients. Client information will be stored securely and reused whenever additional legal documents are required.

Legal Document Template Management

The application shall provide reusable document templates containing placeholders for client information. Templates will eliminate repetitive formatting and ensure consistency across all legal documents.

Document Drafting and Generation

The application shall automatically populate client information into selected legal document templates, allowing users to prepare affidavits and related documents with minimal manual effort.

Work Assignment

The Senior Advocate shall be able to assign document preparation tasks to Junior Advocates. Junior Advocates shall access assigned tasks directly from the application and complete the required documentation.

Document Repository

Every generated legal document shall be permanently archived within the system and linked to the corresponding client profile for future retrieval.

User Management

The application shall support secure role-based access for Senior Advocates and Junior Advocates.

Reports

The application shall generate operational reports related to document generation, work assignments, and client registrations.

1.5 Business Objectives

The implementation of the proposed system aims to achieve the following business objectives.

Improve Office Productivity

The application shall significantly reduce the time required to prepare legal documents by automating repetitive tasks.

Reduce Manual Data Entry

The system shall eliminate repeated typing of client information by storing it permanently within the application.

Improve Document Accuracy

Automatic population of client information shall minimize typographical errors and ensure consistency across all generated documents.

Maintain Permanent Client Records

The application shall preserve complete client information and document history for future reference.

Improve Collaboration

The work assignment feature shall enable the Senior Advocate to distribute work efficiently among Junior Advocates.

Digitize Office Operations

The application shall replace physical registers and manual files with a centralized digital information system.

1.6 Intended Users

The system is designed exclusively for internal office staff.

Only two categories of users are supported.

Administrator (Senior Advocate)

The Administrator is the primary authority responsible for managing the application.

The Administrator shall have permission to:

- Manage office users.
- Register new clients.
- Update client information.
- Create document templates.
- Edit document templates.
- Delete obsolete templates.
- Assign work to Junior Advocates.
- View all generated documents.
- Access reports.
- Configure system settings.

The Administrator has unrestricted access to all system modules.

Junior Advocate

Junior Advocates assist in preparing legal documents.

Junior Advocates shall be able to:

- View assigned work.
- Search registered clients.
- Generate legal documents.
- Edit generated documents.
- Generate printable PDF documents.
- Print completed documents.
- View archived documents based on permissions.

Junior Advocates shall not have permission to:

- Manage users.
- Modify document templates.
- Change system settings.
- Access administrative reports.

1.7 System Benefits

The implementation of NDAOMS provides several advantages over the existing manual process.

Faster Document Preparation

Automated template filling significantly reduces document preparation time.

Improved Accuracy

Automatic insertion of client information minimizes manual errors.

Centralized Information Management

All client records and legal documents are maintained in a single centralized database.

Permanent Digital Archive

Previously generated documents can be retrieved within seconds without searching through physical files.

Better Resource Utilization

The Administrator can effectively distribute work among Junior Advocates.

Standardized Legal Documentation

Reusable templates ensure consistent formatting across all legal documents.

Reduced Paper Dependency

The application minimizes reliance on manual registers and physical document storage.

1.8 Project Limitations (Current Release)

To maintain a manageable scope for the first version, the following features are intentionally excluded:

- Client Portal
- Mobile Application
- Appointment Management
- Court Case Management
- Litigation Tracking
- Payment Gateway
- Online Payments
- Billing & Invoicing
- SMS Notifications
- Email Notifications
- WhatsApp Integration
- AI-Based Legal Drafting
- Digital Signature Integration
- Multi-Branch Office Management

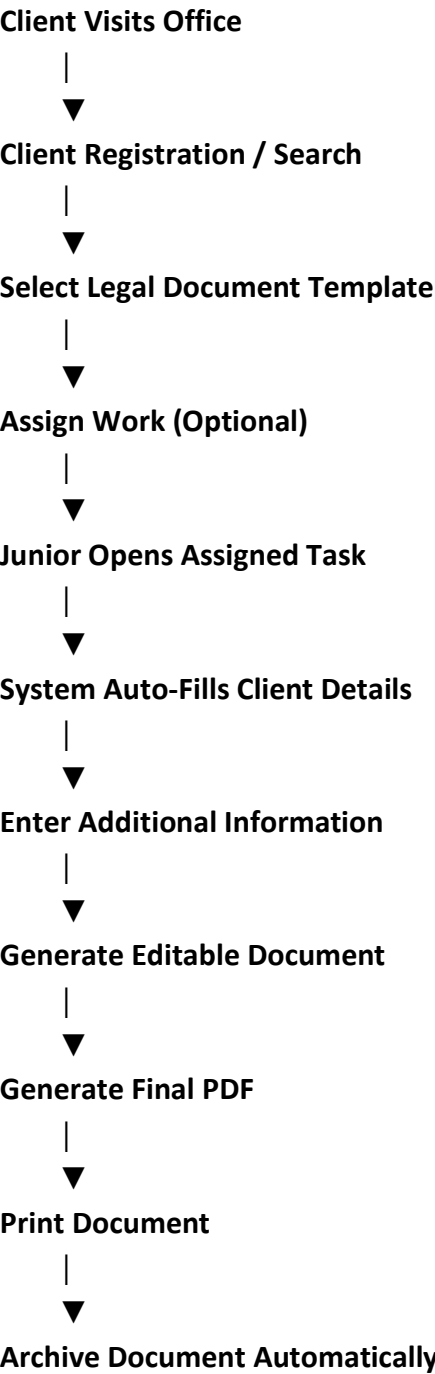
These features may be considered in future releases.

1.9 Definitions and Terminology

Term	Description
Administrator	Senior Advocate with full system privileges
Junior Advocate	User responsible for document preparation
Client	Individual requesting notary services

Template	Predefined legal document format containing placeholders
Placeholder	Dynamic field replaced automatically with client information
Assignment	Task allocated by the Administrator
Generated Document	Final legal document created by the system
Repository	Permanent archive of generated legal documents

1.10 High-Level Business Workflow



Section 2 - Overall Description

2.1 Product Perspective

The **Notary Document Automation and Office Management System (NDAOMS)** is a standalone web-based application developed to automate and simplify the internal operations of a notary advocate office.

The application replaces traditional manual processes such as maintaining physical registers, repeatedly typing client information for every affidavit, and storing printed documents in physical files. Instead, it

provides a centralized digital platform where client information, legal document templates, work assignments, and generated documents are managed efficiently.

The system is designed specifically for internal office operations and is intended to improve document preparation speed, maintain data consistency, and provide quick access to historical records whenever required.

Unlike complete legal practice management software, this system focuses only on **document automation and office management**, making it lightweight, easy to use, and aligned with the client's operational requirements.

2.2 Product Functions

The application consists of several integrated modules that collectively automate the document preparation workflow.

Authentication & User Management

Provides secure login functionality and role-based access for the Administrator and Junior Advocates.

Dashboard

Displays key information such as assigned tasks, registered clients, recently generated documents, and office activity summaries.

Client Management

Maintains a permanent database of client information, allowing advocates to quickly search and reuse client details for future documents.

Document Template Management

Allows the Administrator to create and maintain reusable legal document templates containing dynamic placeholders.

Document Drafting & Generation

Automatically fills client information into the selected template, allows editing of document-specific details, generates printable PDF documents, and archives them.

Work Assignment

Allows the Administrator to assign document preparation tasks to Junior Advocates and monitor the status of assigned work.

Document Archive & Retrieval

Stores every generated document permanently and enables users to search, retrieve, download, and print historical documents.

Reports

Provides summary reports related to client registrations, document generation, work assignments, and office activities.

2.3 User Classes and Characteristics

The system supports only two categories of users.

Administrator (Senior Advocate)

The Administrator is responsible for managing the complete application.

Key responsibilities include:

- Managing users
- Managing clients
- Managing document templates
- Assigning work to Junior Advocates
- Viewing all archived documents
- Generating reports

- Configuring system settings

The Administrator has full access to every module of the application.

Junior Advocate

Junior Advocates assist in preparing legal documents.

They are responsible for:

- Viewing assigned tasks
- Searching client records
- Preparing legal documents
- Editing generated documents
- Generating PDF documents
- Printing documents
- Accessing archived documents according to assigned permissions

Junior Advocates cannot manage users, templates, reports, or system settings.

2.4 Operating Environment

The application will operate in a standard web-based environment.

Component	Technology
Frontend	React.js
UI Framework	Tailwind CSS
Backend	Node.js + Express.js
Database	MongoDB
Authentication	JWT (JSON Web Token)
API Architecture	REST APIs
Document Generation	DOCX Template Engine & PDF Generator
Supported Browsers	Google Chrome, Microsoft Edge, Mozilla Firefox

The system is optimized for desktop and laptop usage within the advocate office.

2.5 Design and Implementation Constraints

The following constraints apply to the first version of the application:

- The system supports only two user roles: Administrator and Junior Advocate.
- All users must authenticate before accessing the application.
- Every generated document must be linked to a registered client.
- Only the Administrator can manage document templates and user accounts.
- The application is intended solely for internal office use.
- The first version will not include payment processing, case management, appointment scheduling, or client-facing functionality.

2.6 Assumptions and Dependencies

The following assumptions have been considered during system design and development:

Assumptions

- Office staff possess basic computer literacy.
- Standard legal document templates are available before deployment.
- Client identity information is collected during registration.
- Users have access to a printer for printing generated documents.

Dependencies

The application depends on:

- MongoDB Database Server
- Node.js Runtime Environment
- Internet or Local Network Connectivity (depending on deployment)
- PDF Generation Library
- Document Template Processing Engine

2.7 System Architecture Overview

The application follows a **three-tier architecture**, separating the user interface, business logic, and database to ensure scalability, maintainability, and security.

Presentation Layer

The frontend is developed using **React.js** and **Tailwind CSS**, providing an intuitive and responsive user interface for both the Administrator and Junior Advocates.

Application Layer

The backend is built using **Node.js** and **Express.js**, which handle authentication, business logic, document generation, work assignments, and communication with the database.

Data Layer

All application data, including users, clients, document templates, assignments, and generated documents, is stored in **MongoDB**.

2.8 System Modules

The application consists of the following primary modules:

Module ID	Module Name
FM-01	Authentication & User Management
FM-02	Dashboard
FM-03	Client Management
FM-04	Document Template Management
FM-05	Document Drafting & Generation
FM-06	Work Assignment
FM-07	Document Archive & Retrieval
FM-08	Reports

These modules work together to automate and streamline the daily operations of the advocate office.

2.9 System Limitations

The current release of the application does not support the following features:

- Client Login Portal
- Mobile Application
- Appointment Scheduling
- Court Case Management
- Litigation Tracking
- Payment Gateway Integration
- Billing & Invoicing
- SMS Notifications
- Email Notifications
- WhatsApp Notifications
- AI-assisted Legal Drafting

- Digital Signature Integration
- Multi-Branch Office Management

These features may be considered for future releases based on business requirements.

Section – 3 Functional Requirements

3.1 Functional Requirements Overview

The functional requirements define the core capabilities of the **Notary Document Automation and Office Management System (NDAOMS)**. These requirements describe the services the application shall provide to its users and specify how the system responds to different user actions.

The application has been divided into functional modules to improve maintainability, simplify development, and ensure scalability. Each module performs a specific business function while interacting with other modules to provide a complete office management solution.

The functional modules included in the current version of the system are:

Module ID	Module Name	Primary User
FM-01	Authentication & User Management	Administrator
FM-02	Dashboard	Administrator, Junior Advocate
FM-03	Client Management	Administrator, Junior Advocate
FM-04	Document Template Management	Administrator
FM-05	Document Drafting & Generation	Administrator, Junior Advocate
FM-06	Work Assignment	Administrator, Junior Advocate
FM-07	Document Archive & Retrieval	Administrator, Junior Advocate
FM-08	Reports & Analytics	Administrator

Each functional module is described in detail in the following sections.

3.2 FM-01 Authentication & User Management

3.2.1 Purpose

The Authentication & User Management module ensures that only authorized users can access the system. It provides secure login, user account management, password management, and role-based access control.

This module protects confidential client information and legal documents from unauthorized access while ensuring that every system activity is associated with an authenticated user.

3.2.2 Description

Every user must authenticate before accessing the application.

The system supports two user roles:

- Administrator (Senior Advocate)
- Junior Advocate

After successful authentication, users are redirected to their respective dashboards according to their assigned roles.

The Administrator is responsible for managing all user accounts within the application.

3.2.3 Actors

Primary Actor

Administrator

Secondary Actor

Junior Advocate

3.2.4 Preconditions

Before using this module:

- The user account must already exist.
- The account must be active.
- The user must possess valid login credentials.
- The application server must be available.

3.2.5 Postconditions

After successful authentication:

- User session is created.
- Dashboard is displayed.
- User permissions are loaded.
- Activity is recorded in the system log.

After logout:

- User session is destroyed.
- Secure login screen is displayed.

3.2.6 Functional Requirements

FR-01 Login

The system shall allow registered users to log in using:

- Username or Email
- Password

Upon successful authentication:

- Create a secure session.
- Redirect the user to the appropriate dashboard.
- Load user permissions.

FR-02 Logout

The system shall allow users to securely log out.

Upon logout:

- Destroy the active session.
- Remove authentication token.
- Redirect user to Login page.

FR-03 Change Password

Authenticated users shall be able to change their password.

The system shall require:

- Current Password
- New Password
- Confirm Password

The system shall verify the current password before updating the new password.

FR-04 User Management

Only the Administrator shall be able to:

- Create Junior Advocate accounts.
- Edit user information.
- Activate user accounts.
- Deactivate user accounts.

- Reset passwords.
- View user list.

Junior Advocates shall not have access to user management.

3.2.7 Inputs

Login

- Username / Email
- Password

User Management

- Full Name
- Email Address
- Mobile Number
- Username
- Password
- Role
- Account Status

3.2.8 Outputs

- Successful Login
- Invalid Credentials
- User Dashboard
- User Created Successfully
- Password Updated
- User Deactivated
- User Activated

3.2.9 Business Rules

BR-01

Each user must have a unique username.

BR-02

Passwords shall be stored using secure encryption.

BR-03

Only Administrators can manage users.

BR-04

Inactive users cannot access the system.

BR-05

Every login attempt shall be recorded in the activity log.

3.2.10 Validation Rules

- Username cannot be blank.
- Password cannot be blank.
- Email must be unique.
- Mobile Number must contain valid digits.
- Password must satisfy minimum security requirements.

3.2.11 Exception Handling

Condition	System Response
Invalid Username	Display appropriate error message
Wrong Password	Display authentication failure

Inactive Account	Prevent login
Server Unavailable	Display system unavailable message

3.2.12 Acceptance Criteria

The module shall be considered complete when:

- Users can log in successfully.
- Sessions are securely managed.
- Passwords are encrypted.
- Only Administrators can manage users.
- Unauthorized users cannot access protected modules.

3.3 FM-02 Dashboard

3.3.1 Purpose

The Dashboard module provides users with a summarized view of office activities immediately after successful login.

It acts as the central navigation point of the application by presenting important statistics, pending work, and shortcuts to frequently used modules.

3.3.2 Description

The dashboard is role-based.

The Administrator dashboard provides complete office statistics, whereas the Junior Advocate dashboard focuses on assigned work and recently generated documents.

The information displayed on the dashboard shall update dynamically based on the latest system data.

3.3.3 Actors

Primary Actors

- Administrator
- Junior Advocate

3.3.4 Preconditions

- User must be authenticated.
- User session must be active.
- Dashboard data must be available.

3.3.5 Postconditions

After accessing the dashboard:

- User can navigate to any permitted module.
- User can view current office statistics.
- User can access assigned work.

3.3.6 Functional Requirements

FR-05 Dashboard Summary

The system shall display a summary of office activities immediately after login.

FR-06 Administrator Dashboard

The Administrator dashboard shall display:

- Total Registered Clients
- Total Generated Documents
- Pending Work Assignments
- Completed Assignments
- Recently Registered Clients
- Recently Generated Documents
- Active Junior Advocates

FR-07 Junior Dashboard

The Junior Advocate dashboard shall display:

- Assigned Work
- Pending Assignments
- Completed Assignments
- Recently Generated Documents
- Personal Work Summary

FR-08 Quick Navigation

The dashboard shall provide shortcuts to:

- Client Management
- Document Generation
- Work Assignment
- Document Archive
- Reports (Administrator only)

3.3.7 Inputs

Dashboard information is automatically retrieved from:

- Client Database
- Document Repository
- Assignment Records
- User Activity

No manual input is required.

3.3.8 Outputs

The dashboard displays:

- Summary Cards
- Pending Tasks
- Recent Activities
- Assigned Work
- Navigation Menu

3.3.9 Business Rules

BR-06

Dashboard information shall be role-specific.

BR-07

Statistics shall be generated using real-time database information.

BR-08

Users shall not view information beyond their assigned permissions.

3.3.10 Validation Rules

- Dashboard shall load only after successful authentication.
- Invalid sessions shall redirect users to the Login page.
- Missing data shall display zero values instead of system errors.

3.3.11 Exception Handling

Condition	System Response
Database unavailable	Display loading/error message
No assignments	Display "No Assigned Work"
No clients registered	Display zero statistics

Session expired	Redirect to Login page
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3.3.12 Acceptance Criteria

The Dashboard module shall be considered complete when:

- Appropriate dashboard loads according to user role.
- Office statistics display correctly.
- Navigation links function correctly.
- Assigned work is displayed accurately.
- Dashboard refreshes using current database information.

3.4 FM-03 Client Management

3.4.1 Purpose

The Client Management module is responsible for maintaining a centralized repository of client information. It allows advocates to register clients, update their information, retrieve previous records, and reuse stored information while generating legal documents.

The objective of this module is to eliminate repeated manual data entry and maintain a permanent history of all clients visiting the office.

3.4.2 Description

Whenever a new client visits the office, the client details shall be recorded in the system. If the client has previously visited the office, the existing record shall be retrieved instead of creating a duplicate record.

Each client record acts as a master profile that can be referenced while generating affidavits and other legal documents.

The module also maintains the history of all documents generated for a particular client.

3.4.3 Actors

Primary Actors

- Administrator
- Junior Advocate

3.4.4 Preconditions

- User must be authenticated.
- User must have permission to access Client Management.

3.4.5 Postconditions

After successful completion of an operation:

- Client information is stored or updated.
- Audit logs are generated.
- Updated information becomes immediately available for document generation.

3.4.6 Functional Requirements

FR-09 Register Client

The system shall allow authorized users to register a new client by entering required information. The system shall automatically generate a unique Client ID.

FR-10 Update Client

Users shall be able to update client information whenever changes occur. The system shall preserve the latest information while maintaining modification history in the audit log.

FR-11 Search Client

The system shall provide advanced search functionality. Search shall be possible using:

- Client Name

- Mobile Number
- Aadhaar Number
- Client ID

FR-12 View Client Profile

Users shall be able to view the complete client profile, including:

- Personal Information
- Identity Details
- Previously Generated Documents
- Registration Date
- Remarks

FR-13 Archive Client Records

Client records shall remain permanently available within the system.

Records shall never be permanently deleted.

3.4.7 Client Information

Each client record shall contain:

- Client ID
- Full Name
- Father's/Husband's Name
- Gender
- Date of Birth (Optional)
- Mobile Number
- Address
- Aadhaar Number
- PAN Number (Optional)
- Occupation (Optional)
- Photograph (Optional)
- Identity Proof
- Registration Date
- Remarks

3.4.8 Business Rules

BR-09

Each client shall have a unique Client ID.

BR-10

Duplicate Aadhaar numbers shall not be permitted.

BR-11

Client records cannot be permanently deleted.

BR-12

Every generated document must be linked to an existing client.

3.4.9 Validation Rules

- Mandatory fields cannot be empty.
- Mobile number shall contain valid digits.
- Aadhaar number must be valid.
- Duplicate client records should be avoided where possible.

3.4.10 Acceptance Criteria

The module shall be accepted when:

- New clients can be registered.

- Existing clients can be searched quickly.
- Client information can be updated.
- Previous documents are accessible from the client profile.
- Duplicate Aadhaar numbers are prevented.

3.5 FM-04 Document Template Management

3.5.1 Purpose

The Document Template Management module enables the office to maintain standardized legal document templates that can be reused for document generation.

This eliminates repetitive formatting and ensures consistency across all legal documents.

3.5.2 Description

Templates are predefined document formats containing placeholders for dynamic information such as the client's name, address, and Aadhaar number.

During document generation, these placeholders are automatically replaced with client information.

Only the Administrator is authorized to manage templates.

3.5.3 Actors

Primary Actor

- Administrator

3.5.4 Functional Requirements

FR-14 Create Template

Administrator shall be able to create new document templates.

FR-15 Edit Template

Administrator shall be able to modify existing templates.

FR-16 Delete Template

Administrator may remove obsolete templates.

Deletion should be logical (inactive) rather than permanent to preserve historical references.

FR-17 Enable / Disable Template

Templates may be activated or deactivated without deleting them.

Inactive templates shall not appear in the document generation module.

FR-18 Preview Template

The system shall provide a preview of templates before use.

3.5.5 Template Placeholders

Templates may include placeholders such as:

- {{ClientName}}
- {{FatherName}}
- {{Address}}
- {{AadhaarNumber}}
- {{Occupation}}
- {{CurrentDate}}

During document generation, the system shall replace these placeholders with actual client data.

3.5.6 Business Rules

BR-13

Every template shall have a unique name.

BR-14

Only Administrators can manage templates.

BR-15

Disabled templates shall not be available for document generation.

3.5.7 Acceptance Criteria

The module shall be accepted when:

- Templates can be created.
- Templates can be edited.
- Placeholders are correctly stored.
- Disabled templates are hidden during document generation.

3.6 FM-05 Document Drafting & Generation

3.6.1 Purpose

The Document Drafting & Generation module is the primary business function of the application. It automates the preparation of legal documents using stored client information and predefined templates.

3.6.2 Description

Users begin by selecting an existing client and an appropriate document template.

The system automatically inserts the client's stored information into the selected template. Users can then enter document-specific details, review the generated content, make necessary edits, generate the final PDF, and print the document.

Every generated document is automatically archived for future reference.

3.6.3 Actors

- Administrator
- Junior Advocate

3.6.4 Preconditions

- User must be logged in.
- Client record must exist.
- Selected template must be active.

3.6.5 Functional Requirements

FR-19 Select Client

The system shall allow users to select a registered client.

FR-20 Select Template

Users shall select the required legal document template.

FR-21 Auto-fill Client Information

The system shall automatically populate template placeholders with stored client information.

FR-22 Enter Document-specific Information

Users shall enter additional information that is unique to the selected document.

Examples include:

- Property Details
- Affidavit Purpose
- Witness Information
- Vehicle Information (if applicable)
- Declaration Details

FR-23 Generate Editable Document

The system shall generate an editable draft for review and modification.

FR-24 Edit Generated Document

Users may modify any part of the generated document before finalizing it.

FR-25 Generate PDF

The system shall convert the final document into PDF format.

FR-26 Print Document

Users shall be able to print the generated PDF directly.

FR-27 Archive Document

The generated document shall automatically be stored in the document repository.

3.6.6 Business Rules

BR-16

Every generated document shall have a unique Document ID.

BR-17

Each document shall be linked to exactly one client.

BR-18

Generated documents shall be archived automatically after PDF generation.

BR-19

Users may edit generated content before printing.

3.6.7 Acceptance Criteria

The module shall be accepted when:

- Client information is automatically populated.
- Placeholders are correctly replaced.
- Users can edit generated documents.
- PDF generation works successfully.
- Generated documents are archived automatically.

3.7 FM-06 Work Assignment

3.7.1 Purpose

The Work Assignment module enables the Administrator to distribute document preparation work among Junior Advocates.

This improves task allocation, workload management, and operational efficiency.

3.7.2 Description

The Administrator may assign a document preparation task to a Junior Advocate.

The assigned Junior Advocate can access the task from the dashboard, prepare the document, and mark the assignment as completed after generating and printing the final document.

The system maintains the current status of each assignment.

3.7.3 Actors

- Administrator
- Junior Advocate

3.7.4 Functional Requirements

FR-28 Assign Work

Administrator shall assign document preparation work to a Junior Advocate.

FR-29 View Assigned Work

Junior Advocates shall view all tasks assigned to them.

FR-30 Update Assignment Status

Junior Advocates shall update the status of assigned work.

Supported statuses include:

- Assigned
- In Progress
- Completed

FR-31 Reassign Work

The Administrator may reassign pending work to another Junior Advocate when required.

FR-32 Assignment History

The system shall maintain a history of all work assignments, including the assigned user, dates, and status changes.

3.7.5 Assignment Information

Each assignment shall contain:

- Assignment ID
- Client Name
- Document Type
- Assigned Junior Advocate
- Assigned By
- Assignment Date
- Due Date (Optional)
- Status
- Remarks

3.7.6 Business Rules

BR-20

Each assignment shall be linked to one client.

BR-21

Each assignment shall be assigned to one Junior Advocate at a time.

BR-22

Completed assignments shall remain available for historical reference.

BR-23

Only the Administrator may assign or reassign work.

3.7.7 Acceptance Criteria

The module shall be accepted when:

- Tasks can be assigned successfully.
- Junior Advocates can access assigned work.
- Assignment status updates correctly.
- Reassignment functions correctly.
- Assignment history is maintained.

3.8 FM-07 Document Archive & Retrieval

3.8.1 Purpose

The **Document Archive & Retrieval** module is responsible for permanently storing all generated legal documents and providing a fast and reliable mechanism to search, retrieve, download, and print previously generated documents.

This module eliminates the dependency on physical document files and enables advocates to retrieve historical documents whenever a client revisits the office.

3.8.2 Description

Every legal document generated by the system is automatically stored in the document repository immediately after PDF generation.

Each archived document is linked to the corresponding client profile, allowing advocates to access complete document history from either the Client Management module or the Archive module. The repository serves as a centralized digital library of all legal documents prepared by the office.

3.8.3 Actors

Primary Actors

- Administrator
- Junior Advocate

3.8.4 Preconditions

Before accessing the archive:

- User must be authenticated.
- User must have permission to access archived documents.
- At least one generated document must exist in the repository.

3.8.5 Postconditions

After successful retrieval:

- Selected document is displayed.
- User may download or print the document.
- Retrieval activity is recorded in the system log (optional).

3.8.6 Functional Requirements

FR-33 Archive Generated Document

The system shall automatically archive every generated document after successful PDF generation.

The archived document shall include:

- Generated PDF
- Document metadata
- Linked client information
- Generation date
- Generated by (User)

FR-34 Search Archived Documents

The system shall provide advanced search functionality to locate archived documents.

Users shall be able to search using one or more of the following parameters:

- Client Name
- Client ID
- Mobile Number
- Aadhaar Number
- Document ID
- Document Type
- Date of Generation
- Assigned Junior Advocate (Administrator only)

FR-35 View Archived Document

Authorized users shall be able to open and view archived documents directly within the system.

FR-36 Download Document

The system shall allow users to download archived documents in PDF format.

FR-37 Print Archived Document

The system shall provide a direct print option for archived documents.

FR-38 View Client Document History

Users shall be able to view all previously generated documents associated with a selected client.

The document history shall display:

- Document ID
- Document Type
- Generation Date
- Generated By
- Print Status (Optional)

3.8.7 Document Metadata

Each archived document shall maintain the following information:

- Document ID
- Client ID
- Client Name
- Document Type
- Template Used
- Generated By
- Assigned User (if applicable)
- Generation Date & Time
- Last Modified Date
- PDF File Location
- Remarks (Optional)

3.8.8 Business Rules

BR-24

Every generated document shall be archived automatically.

BR-25

Archived documents shall remain permanently available unless removed by the Administrator according to organizational policy.

BR-26

Each archived document shall be linked to exactly one client.

BR-27

Only authenticated users shall access archived documents according to their permissions.

BR-28

Archived PDF files shall not be modified directly.
If changes are required, a new document shall be generated.

3.8.9 Validation Rules

- Document ID must exist.
- Search parameters shall return matching records only.
- Users shall not access documents beyond their assigned permissions.
- Missing documents shall display an appropriate message instead of a system error.

3.8.10 Exception Handling

Condition	System Response
Document not found	Display "No matching document found."
Invalid Document ID	Display validation message
Unauthorized Access	Display "Access Denied."
PDF unavailable	Display download error message

3.8.11 Acceptance Criteria

The module shall be accepted when:

- Documents are archived automatically after generation.
- Archived documents are searchable.
- Documents can be viewed successfully.
- PDF download functions correctly.
- Printing archived documents is successful.
- Client document history is displayed accurately.

3.9 FM-08 Reports & Analytics

3.9.1 Purpose

The **Reports & Analytics** module provides the Administrator with operational reports and statistical information regarding office activities.

The purpose of this module is to help monitor office performance, evaluate productivity, and maintain records of daily operations.

3.9.2 Description

The reporting module generates summaries based on data stored within the application.

Reports provide insights into:

- Client registrations
- Document generation
- Work assignments
- Office productivity
- Overall system usage

These reports assist the Administrator in reviewing office performance and identifying workload distribution among Junior Advocates.

3.9.3 Actors

Primary Actor

- Administrator

Junior Advocates do not have access to administrative reports.

3.9.4 Preconditions

- Administrator must be authenticated.
- Report data must exist within the database.

3.9.5 Postconditions

After report generation:

- Requested report is displayed.
- Report may be exported or printed.
- Report generation activity may be logged (optional).

3.9.6 Functional Requirements

FR-39 Daily Document Report

The system shall generate a report showing all legal documents generated on a selected day.

The report shall include:

- Document ID
- Client Name
- Document Type
- Generated By

- Generation Time

FR-40 Monthly Document Report

The system shall generate monthly statistics showing:

- Total documents generated
- Document categories
- Documents generated by each user

FR-41 Client Registration Report

The system shall generate reports showing:

- New client registrations
- Registration dates
- Total registered clients

FR-42 Work Assignment Report

The system shall display assignment statistics including:

- Total assignments
- Pending assignments
- Completed assignments
- Assignments by Junior Advocate

FR-43 User Productivity Report

The system shall generate productivity statistics for Junior Advocates.

The report may include:

- Documents prepared
- Completed assignments
- Average document preparation count

FR-44 Export Reports

The Administrator shall be able to export reports in PDF format.

3.9.7 Report Filters

The reporting module shall support filtering by:

- Date Range
- Month
- Year
- Document Type
- Client Name
- Junior Advocate
- Assignment Status

3.9.8 Business Rules

BR-29

Only the Administrator shall access reports.

BR-30

Reports shall always use the latest available data.

BR-31

Reports shall support date-based filtering.

BR-32

Exported reports shall preserve the same information displayed within the application.

3.9.9 Validation Rules

- Selected date range shall be valid.

- Empty reports shall display "No data available."
- Export shall not proceed if report generation fails.

3.9.10 Exception Handling

Condition	System Response
No report data available	Display "No records found."
Invalid filter selection	Display validation message
Export failed	Display export failure message
Database unavailable	Display system error

3.9.11 Acceptance Criteria

The Reports module shall be considered complete when:

- Reports generate correctly.
- Report filters function properly.
- Report data matches the database.
- PDF export works successfully.
- Only Administrators can access reports.

3.10 Chapter Summary

The Functional Requirements chapter defines all business functionalities required for the **Notary Document Automation and Office Management System**.

The application consists of the following eight functional modules:

Module ID	Module Name
FM-01	Authentication & User Management
FM-02	Dashboard
FM-03	Client Management
FM-04	Document Template Management
FM-05	Document Drafting & Generation
FM-06	Work Assignment
FM-07	Document Archive & Retrieval
FM-08	Reports & Analytics

These modules collectively automate the complete workflow of client management, legal document preparation, work distribution, document archival, and administrative reporting within the advocate office.

Section – 4 External Interface Requirements

4.1 Overview

The External Interface Requirements define how users interact with the **Notary Document Automation and Office Management System (NDAOMS)**. This chapter describes the user interface, software interfaces, hardware interfaces, and communication interfaces required for the proper functioning of the application.

The system is designed as a responsive web application that can be accessed through modern web browsers. It emphasizes simplicity, usability, and consistency to ensure that advocates with basic computer knowledge can operate the application efficiently.

The user interface follows a role-based approach, ensuring that each user only sees the modules and features relevant to their responsibilities.

4.2 User Interface Requirements

The user interface shall provide a clean, intuitive, and responsive experience for both Administrators and Junior Advocates.

The application shall follow a consistent design language across all screens.

General UI Requirements

The system shall provide:

- Responsive web interface
- Simple navigation menu
- Role-based dashboard
- Consistent typography
- Standardized color scheme
- Breadcrumb navigation
- Search functionality
- Confirmation dialogs before critical actions
- Success and error notifications
- Loading indicators during processing

4.2.1 Login Screen

Purpose

Provides secure authentication for authorized users.

Components

- Username / Email field
- Password field
- Show/Hide Password
- Login Button
- Forgot Password (Optional)

Validation

- Username cannot be empty.
- Password cannot be empty.
- Invalid credentials shall display an error message.
- Successful login redirects to the Dashboard.

4.2.2 Dashboard Screen

Purpose

Displays an overview of office activities.

Administrator Dashboard

The dashboard shall display:

- Total Clients
- Total Documents
- Pending Assignments
- Completed Assignments
- Recent Activities
- Quick Navigation Cards

Junior Dashboard

The dashboard shall display:

- Assigned Tasks
- Pending Work
- Completed Work
- Recently Generated Documents

4.2.3 Client Management Screen

Purpose

Manage client records.

Features

- Add Client
- Edit Client
- Search Client
- View Client Profile
- View Document History

Search Filters

- Client Name
- Mobile Number
- Aadhaar Number
- Client ID

4.2.4 Template Management Screen

Purpose

Manage legal document templates.

Features

- Create Template
- Edit Template
- Delete Template
- Preview Template
- Enable/Disable Template

4.2.5 Document Generation Screen

Purpose

Generate legal documents.

Workflow

1. Select Client
2. Select Template
3. Enter Additional Details
4. Generate Document
5. Edit Document
6. Generate PDF
7. Print Document

4.2.6 Work Assignment Screen

Purpose

Assign document preparation work.

Features

Administrator:

- Assign Task
- Reassign Task
- Track Assignment Status

Junior Advocate:

- View Assigned Tasks
- Open Task
- Mark Completed

4.2.7 Document Archive Screen

Purpose

Search and retrieve archived documents.

Features

- Search Documents
- View PDF
- Download PDF
- Print Document
- Client History

4.2.8 Reports Screen

Purpose

Generate operational reports.

Features

- Daily Reports
- Monthly Reports
- Client Reports
- Assignment Reports
- Productivity Reports
- Export PDF

4.3 Software Interface Requirements

The application interacts with several software components to perform its functions.

Interface	Purpose
React.js	User Interface
Node.js	Application Runtime
Express.js	Backend Framework
MongoDB	Data Storage
JWT	Authentication
PDF Generator	PDF Creation
Template Engine	Document Generation

4.4 Hardware Interface Requirements

The application does not require specialized hardware.

Minimum Client Requirements

- Desktop or Laptop
- Dual-Core Processor
- 4 GB RAM
- Keyboard and Mouse
- Printer for document printing

Server Requirements

- Node.js Runtime
- MongoDB Database
- Internet Connectivity
- Storage for archived PDFs

4.5 Communication Interface Requirements

Communication between application components shall occur using secure HTTP-based communication.

Client ↔ Server

- REST APIs
- HTTPS Protocol
- JSON Data Format

Server ↔ Database

- MongoDB Driver

Document Generation

- PDF Generation Library
- Document Template Engine

4.6 User Interface Standards

The application shall follow the following interface standards:

Layout

- Responsive Design
- Fixed Navigation Bar
- Consistent Sidebar
- Card-based Dashboard

Typography

- Sans-serif fonts
- Clear heading hierarchy
- Readable font sizes

Colors

The interface should use a professional color palette.

Suggested colors:

- Primary: Dark Blue
- Secondary: White
- Accent: Light Gray
- Success: Green
- Warning: Orange
- Error: Red

Buttons

Primary Buttons

- Save
- Generate PDF
- Assign
- Login

Secondary Buttons

- Cancel
- Reset
- Preview
- Print

Danger Buttons

- Delete
- Deactivate User

Icons

Use standard icons for:

- Dashboard
- Clients
- Documents
- Templates
- Reports
- Users
- Search
- Print
- Download

4.7 Accessibility Requirements

The interface shall be designed to improve usability for all office users.

The system shall provide:

- Keyboard-accessible navigation.
- High-contrast interface elements.
- Clearly labeled form fields.
- Readable fonts.
- Consistent navigation across screens.
- Meaningful validation messages.
- Confirmation dialogs before destructive actions.

4.8 Error Message Standards

The application shall display meaningful error messages.

Examples include:

Condition	Message
Invalid Login	Invalid username or password.
Duplicate Aadhaar	Client already exists.
Required Field Missing	Please complete all mandatory fields.
Session Expired	Please login again.
Document Not Found	No matching document found.
Server Error	Something went wrong. Please try again later.

4.9 Chapter Summary

This chapter defines the external interfaces of the application, including the user interface, software dependencies, hardware requirements, communication mechanisms, navigation structure, accessibility standards, and error handling.

These requirements ensure that the system provides a consistent, secure, and user-friendly experience while integrating effectively with the selected technology stack.

Section – 5 Database Design

5.1 Overview

The **Database Design** defines how application data will be stored, organized, and managed within the **Notary Document Automation and Office Management System (NDAOMS)**.

The system uses **MongoDB**, a NoSQL document-oriented database, to store application data in the form of collections and documents. MongoDB has been selected due to its flexibility, scalability, and efficient handling of structured and semi-structured data.

The database has been designed to ensure:

- Efficient storage of client records.
- Fast retrieval of historical documents.
- Secure user authentication.
- Centralized management of legal document templates.
- Proper tracking of work assignments.
- Permanent archival of generated legal documents.

Each major module of the application corresponds to one or more MongoDB collections.

5.2 Database Architecture

The database consists of the following primary collections:

Collection Name	Description
-----------------	-------------

Users	Stores Administrator and Junior Advocate details
Clients	Stores client information
Templates	Stores legal document templates
Documents	Stores generated legal documents
Assignments	Stores work assignments
ActivityLogs <i>(Optional)</i>	Stores user activity logs
Reports <i>(Optional)</i>	Stores generated reports if report history is required

The relationships among these collections ensure data consistency and eliminate unnecessary duplication.

5.3 Collection Design

5.3.1 Users Collection

Purpose

Stores authentication and profile information for all users of the application.

Collection Name

users

Fields

Field Name	Data Type	Description
_id	ObjectId	Unique User ID
fullName	String	User's full name
username	String	Unique username
email	String	User email address
password	String	Encrypted password
mobile	String	Contact number
role	String	Admin / Junior
status	Boolean	Active or Inactive
createdAt	Date	Account creation date
updatedAt	Date	Last update

Validation Rules

- Username must be unique.
- Email must be unique.
- Password shall be encrypted.
- Role shall be either Admin or Junior.

5.3.2 Clients Collection

Purpose

Stores permanent records of all clients visiting the office.

Collection Name

clients

Fields

Field Name	Data Type	Description
_id	ObjectId	Client ID
fullName	String	Client Name
fatherName	String	Father's/Husband's Name
gender	String	Gender
dateOfBirth	Date	Optional
mobile	String	Contact Number
address	String	Residential Address
aadhaarNumber	String	Unique Aadhaar Number
panNumber	String	Optional PAN

occupation	String	Optional Occupation
photo	String	Profile Image Path
idProof	String	Identity Document
remarks	String	Additional Notes
createdAt	Date	Registration Date
updatedAt	Date	Last Modified

Validation Rules

- Aadhaar Number must be unique.
- Mobile Number should contain valid digits.
- Client Name is mandatory.

5.3.3 Templates Collection

Purpose

Stores reusable legal document templates.

Collection Name

templates

Fields

Field Name	Data Type	Description
_id	ObjectId	Template ID
templateName	String	Template Name
documentType	String	Affidavit, Declaration, etc.
templateContent	String	Document Content
placeholders	Array	Dynamic Fields
isActive	Boolean	Active Status
createdBy	ObjectId	Admin ID
createdAt	Date	Creation Date
updatedAt	Date	Last Update

Validation Rules

- Template Name must be unique.
- Template Content cannot be empty.

5.3.4 Documents Collection

Purpose

Stores all generated legal documents.

Collection Name

documents

Fields

Field Name	Data Type	Description
_id	ObjectId	Document ID
clientId	ObjectId	Reference to Client
templateId	ObjectId	Reference to Template
generatedBy	ObjectId	User ID
assignmentId	ObjectId	Assignment Reference (Optional)
documentType	String	Document Category
pdfPath	String	PDF Storage Location
generatedContent	String	Final Document Content
generatedDate	Date	Date Generated
lastModified	Date	Last Modified

Validation Rules

- Client must exist.
- Template must exist.

- PDF shall be generated before archival.

5.3.5 Assignments Collection

Purpose

Stores work assignments allocated to Junior Advocates.

Collection Name

assignments

Fields

Field Name	Data Type	Description
_id	ObjectId	Assignment ID
clientId	ObjectId	Client Reference
assignedTo	ObjectId	Junior Advocate
assignedBy	ObjectId	Administrator
documentType	String	Required Document
status	String	Assigned / In Progress / Completed
remarks	String	Additional Notes
assignedDate	Date	Assignment Date
completedDate	Date	Completion Date

Validation Rules

- Assignment must reference a valid client.
- Assignment must reference a valid Junior Advocate.
- Status shall be one of:
 - Assigned
 - In Progress
 - Completed

5.3.6 Activity Logs Collection (Optional)

Purpose

Maintains records of important user activities.

Collection Name

activity_logs

Fields

- User ID
- Activity
- Module
- Timestamp
- IP Address (Optional)

Examples:

- User Login
- User Logout
- Client Registration
- Document Generated
- Assignment Created

5.4 Collection Relationships

The following relationships exist among the collections.

Users → Assignments

One Administrator can assign multiple work assignments.

One Junior Advocate can receive multiple assignments.

Relationship: One-to-Many

Clients → Documents

One client may have multiple generated legal documents.

Relationship: One-to-Many

Templates → Documents

One template may generate multiple legal documents.

Relationship: One-to-Many

Assignments → Documents

One assignment may produce one generated document.

Relationship: One-to-One (Current Version)

Users → Documents

One user may generate multiple documents.

Relationship: One-to-Many

5.5 Database Relationship Summary

Parent Collection	Child Collection	Relationship
Users	Assignments	One-to-Many
Users	Documents	One-to-Many
Clients	Documents	One-to-Many
Templates	Documents	One-to-Many
Assignments	Documents	One-to-One

5.6 Data Integrity Rules

The following integrity rules shall be enforced by the application:

- Every document shall reference an existing client.
- Every assignment shall reference an existing Junior Advocate.
- Every template shall have a unique name.
- Every client shall have a unique Aadhaar Number.
- Usernames shall be unique.
- Deleted templates shall not remove previously generated documents.
- Archived documents shall remain linked to their respective clients.

5.7 Data Backup and Recovery

To ensure data availability and prevent accidental data loss, the following backup strategies shall be implemented:

- Periodic backup of the MongoDB database.
- Backup of generated PDF documents.
- Recovery procedures for restoring database and document files.
- Verification of backup integrity after scheduled backups.

These measures help maintain business continuity in the event of hardware failure, accidental deletion, or software issues.

5.8 Database Security

The database shall implement the following security measures:

- Passwords stored using secure hashing algorithms.
- Role-based database access.
- Secure API communication using HTTPS.
- JWT authentication for user sessions.
- Input validation to prevent injection attacks.

- Regular database backups.
- Restricted direct database access to authorized administrators.

5.9 Chapter Summary

This chapter describes the database structure of the **Notary Document Automation and Office Management System**, including MongoDB collections, field definitions, relationships, integrity rules, backup strategies, and security measures.

The proposed database design provides a scalable and maintainable foundation for managing users, clients, templates, assignments, and generated legal documents while ensuring data consistency, security, and efficient retrieval.

Section – 6 Non-Functional Requirements

6.1 Overview

Non-Functional Requirements (NFRs) define the quality standards and operational characteristics of the **Notary Document Automation and Office Management System (NDAOMS)**. Unlike functional requirements, which describe specific system features, non-functional requirements specify how the system should behave in terms of performance, security, reliability, usability, scalability, and maintainability.

The application shall be designed to provide a secure, responsive, and user-friendly experience while ensuring data integrity and efficient document management. These requirements serve as measurable quality benchmarks throughout the software development lifecycle.

6.2 Performance Requirements

The system shall deliver acceptable performance under normal office operations.

NFR-01 Response Time

- User login should complete within **3 seconds** under normal network conditions.
- Search operations for clients and documents should return results within **2 seconds**.
- Dashboard statistics should load within **3 seconds** after successful login.
- Page navigation should complete within **2 seconds**.

NFR-02 Document Generation Performance

- The system shall generate an editable legal document within **5 seconds**.
- PDF generation should complete within **10 seconds**, depending on document size.
- Printing shall be initiated immediately after user confirmation.

NFR-03 Concurrent Users

The system shall support multiple authenticated users working simultaneously without affecting normal performance.

Expected concurrent users:

- Administrator: 1
- Junior Advocates: Up to 10 (configurable)

NFR-04 Database Performance

The database shall efficiently manage:

- Thousands of client records.
- Thousands of archived legal documents.
- Multiple work assignments.
- High-speed search operations.

Database indexing should be applied to frequently searched fields such as:

- Client Name

- Aadhaar Number
- Mobile Number
- Document ID

6.3 Security Requirements

The application stores confidential legal information and therefore requires strong security controls.

NFR-05 Authentication

Only authenticated users shall access the application.

Every user shall log in using valid credentials.

NFR-06 Authorization

Role-Based Access Control (RBAC) shall be implemented.

Permissions shall be assigned according to user roles.

Administrator:

- Full system access

Junior Advocate:

- Limited access based on assigned responsibilities

NFR-07 Password Security

Passwords shall:

- Never be stored in plain text.
- Be securely hashed before storage.
- Be validated during authentication.

NFR-08 Session Management

The system shall:

- Generate secure authentication tokens.
- Expire inactive sessions automatically.
- Destroy user sessions after logout.

NFR-09 Data Protection

The application shall protect confidential client information.

Only authorized users shall access client records and archived legal documents.

NFR-10 API Security

REST APIs shall implement:

- JWT Authentication
- Authorization Middleware
- Input Validation
- Request Authentication

6.4 Reliability Requirements

The system shall provide reliable operation during daily office activities.

NFR-11 Data Integrity

The application shall maintain data consistency throughout all operations.

No generated document shall exist without an associated client.

NFR-12 Error Recovery

In case of unexpected failures:

- Meaningful error messages shall be displayed.
- Data corruption shall be prevented.
- Users shall be able to continue working after recovery.

NFR-13 Backup Reliability

Regular database backups shall be maintained to minimize the risk of data loss.

6.5 Availability Requirements

The application shall remain available during normal office working hours.

Expected availability:

- **99% uptime** during operational hours, excluding planned maintenance.

Scheduled maintenance should be communicated in advance.

6.6 Usability Requirements

The system shall be easy to learn and operate.

NFR-14 User-Friendly Interface

The interface shall provide:

- Simple navigation.
- Clearly labeled menus.
- Consistent layouts.
- Easy-to-understand forms.

NFR-15 Minimal Training

A new office employee should be able to perform basic operations after a short orientation.

NFR-16 Form Validation

Forms shall provide immediate validation feedback for:

- Required fields
- Invalid inputs
- Duplicate records
- Missing information

NFR-17 Error Messages

Validation and system messages shall be clear, concise, and informative.

Technical errors should not expose internal implementation details.

6.7 Maintainability Requirements

The application shall be designed for easy maintenance and future enhancements.

NFR-18 Modular Architecture

The system shall follow a modular architecture where each functional module is independent and loosely coupled.

NFR-19 Code Maintainability

The application shall follow:

- Clean coding practices.
- Consistent naming conventions.
- Proper code documentation.
- Reusable components.

NFR-20 Configuration Management

Configuration values such as database connections, API endpoints, and environment variables shall be managed separately from application source code.

6.8 Scalability Requirements

The application shall support future growth without requiring major architectural changes.

Future scalability should include:

- Increased client records.

- Additional Junior Advocates.
- More document templates.
- Increased document archive.
- Additional reports.
- Future feature expansion.

6.9 Compatibility Requirements

The application shall operate across commonly used web browsers.

Supported browsers include:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox

The system shall function correctly on desktop and laptop devices with standard screen resolutions.

6.10 Portability Requirements

The application shall be deployable in different environments with minimal configuration changes.

Supported deployment options include:

- Local office server
- Cloud server
- Virtual Private Server (VPS)

Configuration files should allow environment-specific settings without requiring source code modification.

6.11 Backup and Recovery Requirements

To protect important legal records, the application shall support data backup and recovery procedures.

Backup requirements include:

- Periodic database backups.
- Backup of generated PDF documents.
- Secure storage of backup files.
- Restoration capability in case of accidental data loss or system failure.

Backup procedures shall be tested periodically to ensure data can be successfully restored.

6.12 Logging and Monitoring Requirements

The system shall maintain activity logs for important operations to support auditing and troubleshooting.

The following events should be logged:

- User login
- User logout
- Client registration
- Client updates
- Document generation
- Work assignment
- Template creation or modification
- System errors (without exposing sensitive data)

Logs should include:

- User ID
- Timestamp
- Action performed
- Module affected

6.13 Legal and Compliance Requirements

The application shall comply with general software engineering best practices for protecting sensitive user information.

The system shall:

- Restrict unauthorized access to client records.
- Preserve document integrity.
- Maintain audit information for administrative activities.
- Support secure storage of confidential legal data.

6.14 Non-Functional Requirement Summary

Requirement Category	Requirement ID
Performance	NFR-01 to NFR-04
Security	NFR-05 to NFR-10
Reliability	NFR-11 to NFR-13
Availability	NFR-14
Usability	NFR-15 to NFR-18
Maintainability	NFR-19 to NFR-20
Scalability	NFR-21
Compatibility	NFR-22
Portability	NFR-23
Backup & Recovery	NFR-24
Logging & Monitoring	NFR-25
Legal & Compliance	NFR-26

Note: The numbering above groups related requirements. In the final SRS, you may either keep the detailed numbering used in each subsection or renumber them sequentially from **NFR-01** onward for consistency.

6.15 Chapter Summary

This chapter defines the non-functional characteristics of the **Notary Document Automation and Office Management System**, including performance, security, reliability, usability, maintainability, scalability, compatibility, portability, backup and recovery, logging, and compliance requirements.

These requirements ensure that the system not only provides the required functionality but also operates efficiently, securely, and reliably in a real-world office environment.

Section – 7 Business Rules

7.1 Overview

Business Rules define the policies, constraints, and operational guidelines that govern the behavior of the **Notary Document Automation and Office Management System (NDAOMS)**. These rules ensure that the application operates according to the business processes followed in a notary advocate office.

The rules described in this chapter are independent of the implementation technology and represent the office's operational policies. They guide software development, testing, and future maintenance by ensuring that the application consistently enforces the agreed business processes.

7.2 User Management Rules

BR-01 User Roles

The system shall support only two user roles:

- Administrator (Senior Advocate)
- Junior Advocate

No additional user roles shall be available in the current version.

BR-02 User Authentication

Every user must successfully authenticate before accessing any system module.
Unauthorized users shall not access application resources.

BR-03 Unique User Accounts

Each user account shall have:

- Unique Username
- Unique Email Address

Duplicate user accounts shall not be permitted.

BR-04 Password Security

Passwords shall never be stored in plain text.
The system shall securely hash passwords before storing them in the database.

BR-05 User Status

Only active users shall be allowed to access the application.
Inactive users shall be prevented from logging in until reactivated by the Administrator.

7.3 Client Management Rules

BR-06 Client Registration

Every client shall be registered only once in the system.
If a matching client already exists, the existing record shall be reused instead of creating a duplicate.

BR-07 Unique Client Identification

Each client shall have:

- Unique Client ID
- Unique Aadhaar Number

These identifiers shall uniquely identify the client within the application.

BR-08 Mandatory Client Information

The following information shall be mandatory during client registration:

- Full Name
- Father's/Husband's Name
- Mobile Number
- Address
- Aadhaar Number

Optional fields may be left blank.

BR-09 Client Record Retention

Client records shall not be permanently deleted from the database.
If required, records may be marked as inactive or archived while preserving historical information.

BR-10 Client History

All legal documents generated for a client shall remain permanently linked to that client's profile.

7.4 Document Template Rules

BR-11 Template Ownership

Only the Administrator shall create, edit, enable, disable, or delete document templates.
Junior Advocates shall not modify templates.

BR-12 Template Availability

Only active templates shall be available for document generation.
Inactive templates shall remain stored but hidden from users.

BR-13 Template Identification

Every template shall have a unique name.

Two templates with the same name shall not exist simultaneously.

BR-14 Placeholder Replacement

Every placeholder within a template shall be automatically replaced with the corresponding client or document information during document generation.

7.5 Document Generation Rules

BR-15 Registered Client Requirement

Legal documents shall only be generated for registered clients.
Users shall not generate documents for unregistered clients.

BR-16 Active Template Requirement

Only active document templates shall be used for document generation.

BR-17 Editable Draft

After automatic data population, users shall be allowed to edit the generated document before producing the final PDF.

BR-18 Unique Document Identification

Every generated legal document shall receive a unique Document ID.

BR-19 Automatic Archiving

Every successfully generated PDF shall be archived automatically within the system.
No manual archiving process shall be required.

BR-20 Document Ownership

Each generated document shall be linked to:

- One Client
- One Template
- One User who generated it

7.6 Work Assignment Rules

BR-21 Assignment Authority

Only the Administrator shall assign or reassign document preparation work.

BR-22 Single Responsibility

Each assignment shall be assigned to only one Junior Advocate at a time.

BR-23 Assignment Status

Assignments shall follow one of the following statuses:

- Assigned
- In Progress
- Completed

No additional statuses shall be supported in the first release.

BR-24 Completed Assignments

Completed assignments shall remain available for historical reference and reporting.

BR-25 Document Generation

Junior Advocates shall generate and print documents directly after completing assigned work.
No approval from the Administrator shall be required before printing.

7.7 Document Archive Rules

BR-26 Automatic Storage

All generated documents shall be automatically stored in the document repository.

BR-27 Permanent Storage

Archived documents shall remain permanently available unless removed according to organizational data retention policies.

BR-28 Search Availability

Users shall be able to search archived documents using available search filters.

BR-29 Document Integrity

Archived PDF files shall not be modified after generation.

If corrections are required, a new document shall be generated and archived.

7.8 Reporting Rules

BR-30 Administrator Access

Only the Administrator shall access the reporting module.

BR-31 Report Accuracy

Reports shall always be generated using the latest available data from the database.

BR-32 Report Filters

Reports shall support filtering by:

- Date Range
- Month
- Year
- Client
- Document Type
- Junior Advocate

BR-33 Export Capability

Reports shall be exportable in PDF format.

7.9 Data Integrity Rules

BR-34 Referential Integrity

Every document shall reference an existing:

- Client
- Template
- User

BR-35 Assignment Integrity

Every assignment shall reference:

- One Client
- One Junior Advocate
- One Administrator

BR-36 Duplicate Prevention

The system shall prevent duplicate records for:

- Usernames
- Email Addresses
- Aadhaar Numbers
- Template Names

BR-37 Audit Trail

Important operations shall be recorded in the activity log.

These operations include:

- Login
- Logout

- Client Registration
- Client Updates
- Template Management
- Document Generation
- Work Assignment

7.10 Security Rules

BR-38 Role-Based Access Control

System permissions shall be determined according to user roles.

BR-39 Session Security

Only authenticated users with active sessions shall access protected resources.

BR-40 Confidentiality

Client information and legal documents shall only be accessible to authorized users.

7.11 Exception Handling Rules

BR-41 Validation Errors

The system shall prevent saving incomplete or invalid information.

Appropriate validation messages shall be displayed to the user.

BR-42 System Failures

Unexpected system failures shall not result in partial or inconsistent data being stored in the database.

Transactions should either complete successfully or fail safely.

BR-43 Recovery

Following a recoverable failure, the application shall allow users to continue working without data corruption.

7.12 Business Rule Summary

Category	Business Rule IDs
User Management	BR-01 to BR-05
Client Management	BR-06 to BR-10
Template Management	BR-11 to BR-14
Document Generation	BR-15 to BR-20
Work Assignment	BR-21 to BR-25
Document Archive	BR-26 to BR-29
Reporting	BR-30 to BR-33
Data Integrity	BR-34 to BR-37
Security	BR-38 to BR-40
Exception Handling	BR-41 to BR-43

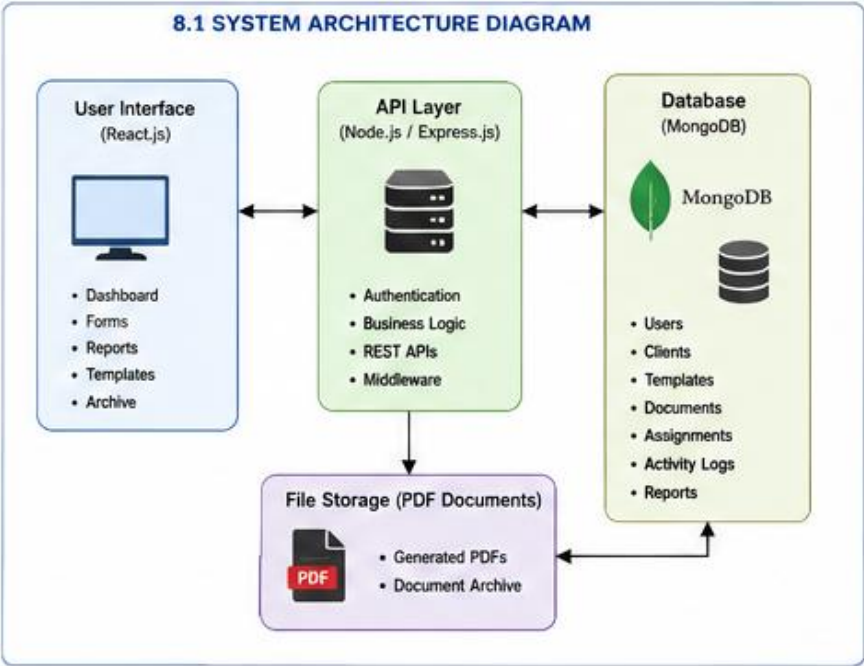
7.13 Chapter Summary

This chapter defines the operational policies that govern the **Notary Document Automation and Office Management System**. These business rules ensure that client information, legal documents, user accounts, work assignments, and reports are managed consistently and securely according to the office's established procedures.

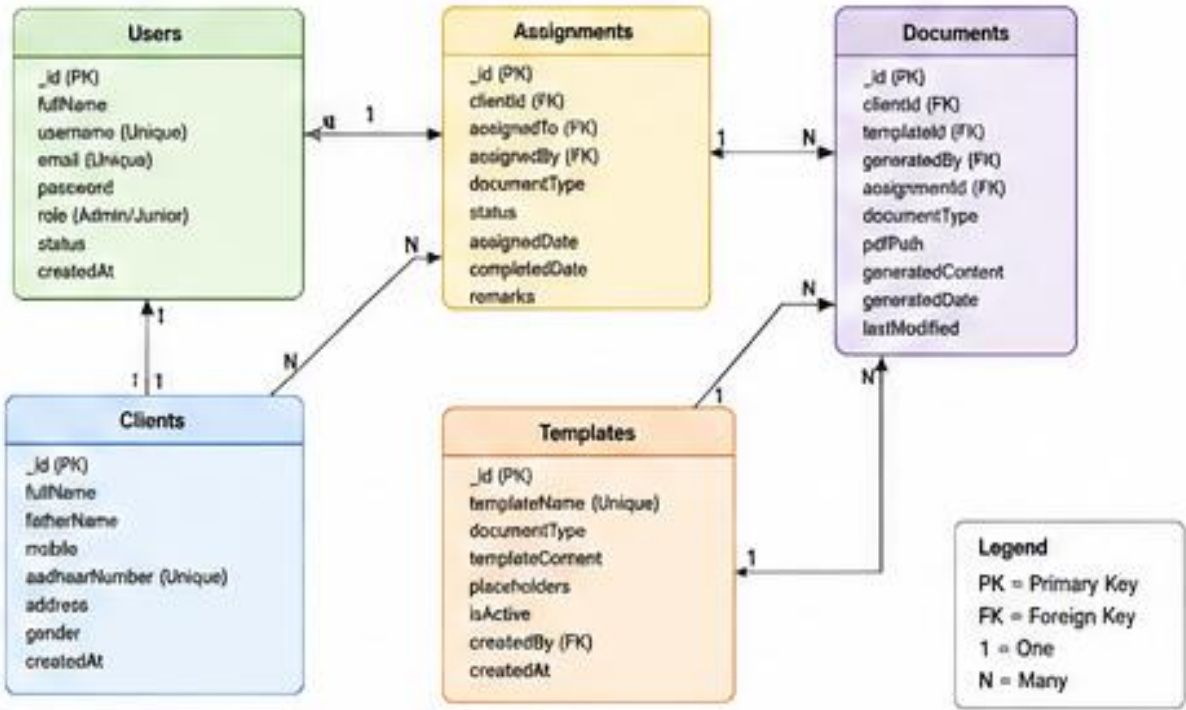
The rules presented in this chapter provide a single reference point for developers, testers, and stakeholders, helping to ensure that the implemented system accurately reflects the client's business processes.

Section – 8 System Designs & Diagrams

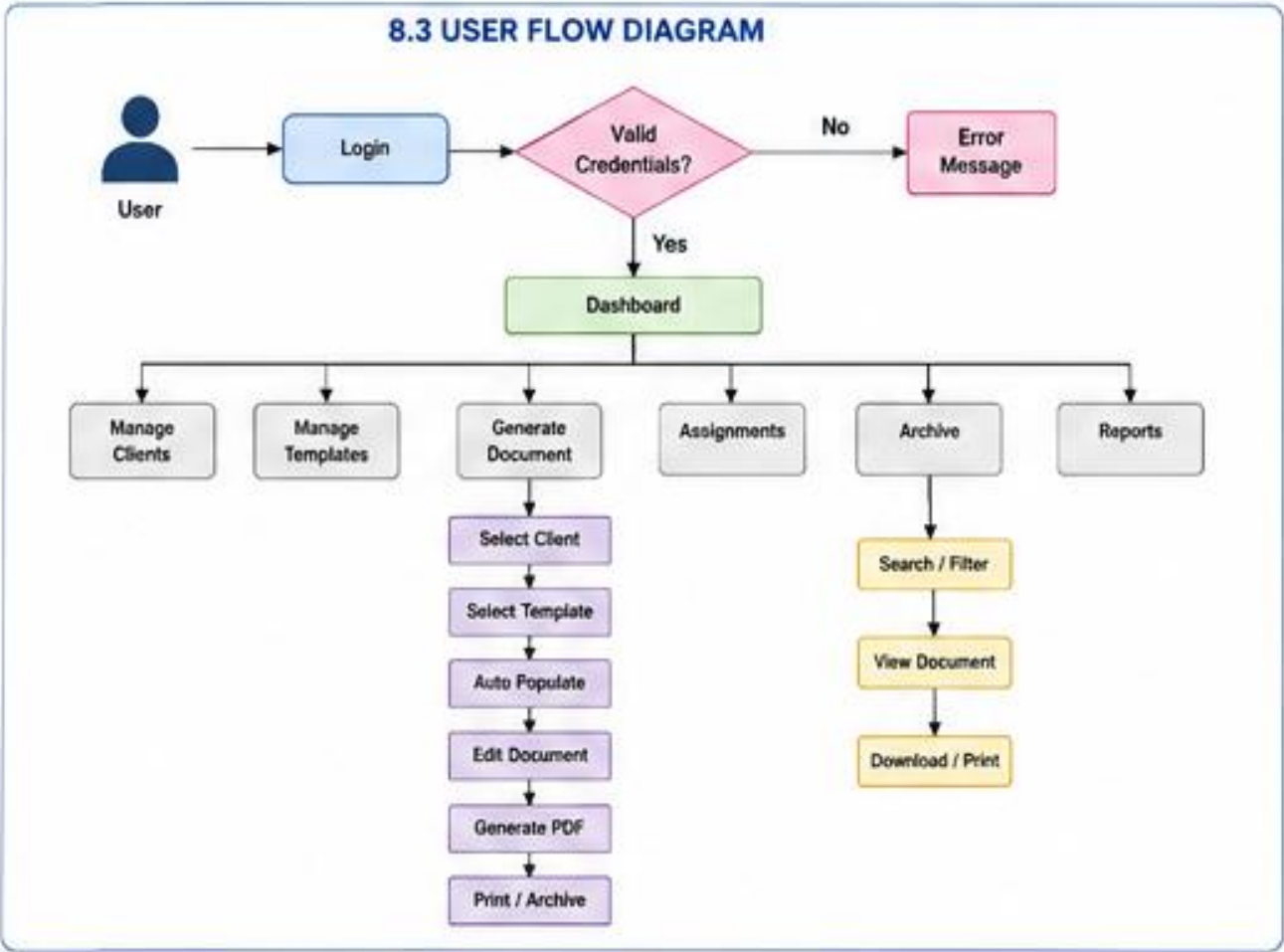
8.1 System Architecture Diagram



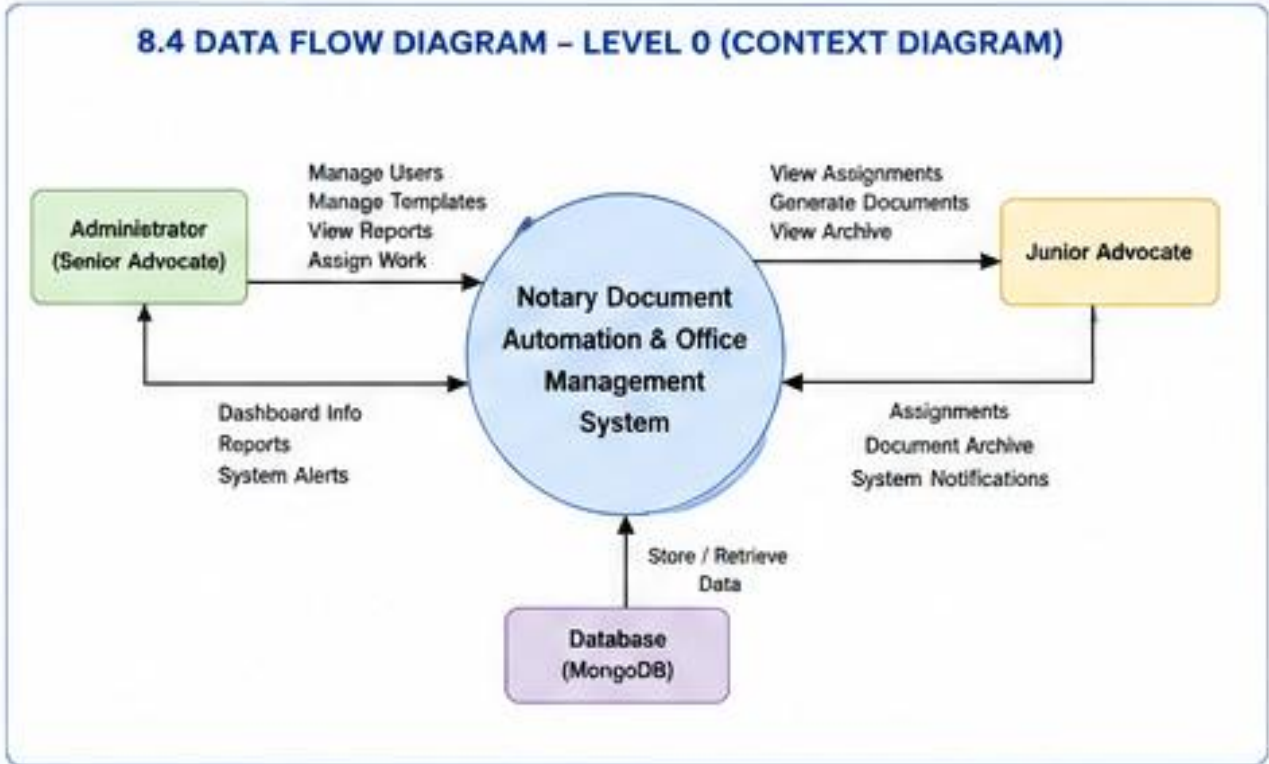
8.2 ER Diagram



8.3 User Flow Diagram



8.4 Data Flow Diagram – Level 0



8.5 Data Flow Diagram – Level 1

